8 The Chemical Potential

In this section we address some unfinished business: that unlike the pairs of extensive and intensive variables S and T, and V and p, the number of particles, N, has been a bystander.

There has been no consideration of a system exchanging particles with its environment, unlike energy or volume. This section explores the extension to this circumstance.

The original examples were chemical reactions where the numbers of a given molecule could change. For example within water there is an equilibrium[1] between water molecules and ions:

$$\mathrm{H_2O} \rightleftharpoons \mathrm{H^+} + \mathrm{OH^-} \ .$$

Later the idea was used in quantum systems, for example electrons entering a sample through electrical contacts (so the systems are spatially distinct – unlike the chemical case). But there is also a convenience to allowing particle number to vary. In the same manner that fixed energy is harder to deal with than a system in contact with a heat bath, the same is true for quantum gases in terms of contact with a "particle bath" being easier. So this section is a prelude to the discussion of quantum gases and liquids: from electrons in solids to ultracold gases to neutron stars.

8.1 Thermodynamics: macroscopic particle exchange

Let us start with the internal energy, $U(S, V, N)$. Then we may write, assuming for large N we may regard N as a continuous parameter:

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_{V,N} \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_{N,S} \mathrm{d}V + \left(\frac{\partial U}{\partial N}\right)_{S,V} \mathrm{d}N \tag{8.1}$$

$$= T\,\mathrm{d}S - p\,\mathrm{d}V + \mu\,\mathrm{d}N \ , \tag{8.2}$$

introducing the *chemical potential*, μ, as the conjugate, intensive, variable to N;

$$\mu = \left(\frac{\partial U}{\partial N}\right)_{S,V} = U(S, V, N) - U(S, V, N-1) \ ,$$

where the latter expression respects the integer nature of N.

[1] Almost all water molecules are neutral, the concentration of ions is roughly $10^{-4}\,\mathrm{mol\,m^{-3}}$. But sometimes equilibrium is difficult to attain as in $2\,\mathrm{H_2} + \mathrm{O_2} \rightleftharpoons 2\,\mathrm{H_2O}$, where it is difficult to be in a situation where all three constituents are in significant concentrations. If all the molecules are $\mathrm{H_2}$ and $\mathrm{O_2}$ the system is metastable, complex and explosive! (See p 833 Atkins *Physical Chemistry* 8th edition WH Freeman (2006).)

It is useful (when we turn to statistical mechanics with particle exchange) to rewrite these equations with entropy, $S(U,V,N)$, as the prime focus (anticipating a parallel with our initial discussion of exchange of energy in section 3.3):

$$\mathrm{d}S = \frac{1}{T}\,\mathrm{d}U + \frac{p}{T}\,\mathrm{d}V - \frac{\mu}{T}\,\mathrm{d}N \tag{8.3}$$

$$= \left(\frac{\partial S}{\partial U}\right)_{V,N}\mathrm{d}U + \left(\frac{\partial S}{\partial V}\right)_{N,U}\mathrm{d}V + \left(\frac{\partial S}{\partial N}\right)_{U,V}\mathrm{d}N\ , \tag{8.4}$$

where we may read off the partial derivatives of S, by comparison of the coefficients of the differentials in Eq. (8.3) and Eq. (8.4), especially

$$\frac{\mu}{T} = -\left(\frac{\partial S}{\partial N}\right)_{U,V}\ . \tag{8.5}$$

Finally we now deduce a simple relationship between μ and the Gibbs free energy, $G(T,p,N)$.

Imagine we have two copies of a system with extensive variables U, S, V and N. We could "glue them" together, allowing them to exchange energy, volume and particle number.

We then expect we have *one* combined system, with extensive variables $2U$, $2S$, $2V$ and $2N$. I.e

$$2U(S,V,N) = U(2S,2V,2N)\ .$$

Let us consider a general change of size: $V \to \lambda V$, where $\lambda = 2$ in the example above. Then :

$$\lambda U(S,V,N) = U(\lambda S,\lambda V,\lambda N)\ .$$

We now differentiate both sides of this equation with respect to λ. The left hand side is straightforward:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}[\lambda U(S,V,N)] = U(S,V,N)\ .$$

The righthand side is an exercise in the formulation end of sec 4.1 of semester 2 Y1 Maths.

Let $x_1(\lambda,S) = \lambda S$, $x_2(\lambda,V) = \lambda V$ and $x_3(\lambda,N) = \lambda N$. Then,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}U(x_1(\lambda,S),x_2(\lambda,V),x_3(\lambda,N)) = \left.\frac{\partial U}{\partial x_1}\right|_{x_1=\lambda S}\frac{\partial x_1}{\partial \lambda} + \left.\frac{\partial U}{\partial x_2}\right|_{x_2=\lambda V}\frac{\partial x_2}{\partial \lambda} + \left.\frac{\partial U}{\partial x_3}\right|_{x_3=\lambda N}\frac{\partial x_3}{\partial \lambda}$$

$$U = \frac{\partial U}{\partial \lambda S}\frac{\partial \lambda S}{\partial \lambda} + \frac{\partial U}{\partial \lambda V}\frac{\partial \lambda V}{\partial \lambda} + \frac{\partial U}{\partial \lambda N}\frac{\partial \lambda N}{\partial \lambda}$$

(Set $\lambda = 1$.) $$= \frac{\partial U}{\partial S}S + \frac{\partial U}{\partial V}V + \frac{\partial U}{\partial N}N$$

(Use Eq. (8.1), Eq. (8.2).) $$U = TS - pV + \mu N \tag{8.6}$$

$$\Rightarrow \quad U - TS + pV = \mu N \tag{8.7}$$

$$\Rightarrow \quad G = \mu N \tag{8.8}$$

$$\mu = \frac{G}{N}\ .$$

So μ is the Gibbs free energy per particle. Why G is selected is mysterious physically. For the other thermodynamic functions, there is a differential relation, but not an overall linear relation, eg

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V} \neq \frac{F}{N} \ ,$$

in general.

We may also rewrite (and sometimes "simplify") the other thermodynamic potentials in a similar manner, using Eq. (8.6), including Ω, which we meet presently:

$$\begin{aligned} F &= U - TS &&= -pV + \mu N \\ H &= U + pV &&= TS - \mu N \\ \Omega &= U - TS - \mu N &&= -pV \ . \end{aligned}$$

These relations can be useful in statistical mechanics as well, e.g. the last line to deduce p if we know Ω and fix V.

8.1.1 The Gibbs-Duhem relation – no thermodynamic superpotential – non-examinable

This subsection shows there is no thermodynamic potential, $J(T, p, \mu)$, which only depends on the intensive variables, as they are not all independent.

Remember

$$\begin{aligned} & & G(T,p,N) &= U + pV - TS \\ & \text{(So we expect:)} & J(T,p,\mu) &= G(T,p,N) - \mu N \\ & \text{(But } G = \mu N\text{:)} \quad \Rightarrow & J(T,p,\mu) &= \mu N - \mu N \\ & & &= 0 \ . \end{aligned}$$

So there is no thermodynamic potential with only intensive variables, although when N is finite one can have all three intensive variables *Thermodynamics of small systems*, Terrell L. Hill, Dover (1994), Chapter 10.

This is consistent with the three intensive variables being related. There are at least two ways to deduce this.

Firstly consider, for example, the three partial derivatives of the entropy, $S(U,V,N)$:

$$\begin{aligned}\frac{1}{T(U,V,N)} &= \left(\frac{\partial S}{\partial U}\right)_{V,N} \\ \frac{p(U,V,N)}{T(U,V,N)} &= \left(\frac{\partial S}{\partial V}\right)_{N,U} \\ -\frac{\mu(U,V,N)}{T(U,V,N)} &= \left(\frac{\partial S}{\partial N}\right)_{U,V} .\end{aligned}$$

In principle, one can eliminate U, V, and N from these three equations to leave a relation between T, p and μ.

Secondly we may deduce a differential relation between the intensive variables:

$$\begin{aligned} U - TS + pV - \mu N &= 0 \\ \Rightarrow \quad \mathrm{d}U - \mathrm{d}(TS) + \mathrm{d}(pV) - \mathrm{d}(\mu N) &= 0 \\ \Rightarrow \quad -S\,\mathrm{d}T + V\,\mathrm{d}p - N\,\mathrm{d}\mu &= 0 \quad \text{(Using the First Law.)} .\end{aligned}$$

This is the *Gibbs-Duhem* relation, a differential relation between T, p and μ, stating they cannot all be varied separately.

8.2 Statistical mechanics with particle exchange

Here we follow our previous development where we considered two systems exchanging energy and then the case where one had a much larger thermal capacity than the other, forming a bath which determined the temperature. We now perform the analogous derivations but allowing particle exchange *as well as* energy exchange.[2] Note that we assume both energy and particle number are conserved here.

8.2.1 Two systems exchanging particles

This should be compared with section 3.3. Consider two systems labelled 1 and 2 which are statistically independent so that the number of configurations, $W^{\text{tot}}(E,N)$, with total energy, $E = E_1 + E_2$, and particle number, $N = N_1 + N_2$, (note the conservation of total particle number) is the product of the number of configurations of the two systems:

$$W^{\text{tot}}(E,N) = W_1(E_1,N_1)\underbrace{W_2(E-E_1,N-N_1)}_{E_2 = E - E_1 \text{ etc.}} .$$

[2] It does not seem easy to consider particle exchange *without* energy exchange, as the particles could then not carry energy and thus have no observable effect. Whether the effects of exchanged entropy with no energy exchange could be formulated is not clear.

Let us now maximise the total number of configurations both with respect to E_1 and N_1. The energy maximisation is the same as in section 3.3 so we will not repeat – it leads to the equality of the temperatures

$$\frac{1}{T_1} = \frac{1}{T_2} \ .$$

With regard to maximising wrt N_1, using $'$ as being the derivative with respect to the argument of the function:

$$\begin{aligned}
& \frac{\partial W^{\text{tot}}}{\partial N_1} = 0 = W_1'(N_1)W_2(N-N_1) + W_1(N_1)(-1)W_2'(N-N_1) \\
& \Rightarrow \quad \frac{W_1'}{W_1} = \frac{W_2'}{W_2} \\
\Rightarrow \quad & \frac{\partial}{\partial N_1}\ln W_1 = \frac{\partial}{\partial N_2}\ln W_2 \quad \text{(Use } N_2 = N - N_1.\text{)} \\
& \frac{\partial S_1}{\partial N_1} = \frac{\partial S_2}{\partial N_2} \quad \text{(Multiplying by } k_{\text{B}}.\text{)} \\
\text{(Use Eq. (8.5).)} \quad \Rightarrow \quad & -\frac{\mu_1}{T_1} = -\frac{\mu_2}{T_2} \\
\Rightarrow \quad & \mu_1 = \mu_2 \ ,
\end{aligned}$$

as we have already shown independently that energy exchange ensures the temperatures are equal.

Thus: the chemical potentials of systems exchanging particles are equal in equilibrium – with maximum entropy.

8.2.2 Particle bath in contact with system

We treat the numbers of particles as a continuous variable again. This should be accurate when $N \to \infty$. We follow the derivation of section 3.4, where the role of a heat bath was determined. Here we need to follow both E and N.

Then if $W_{\text{s}}(E_{\text{s}}, N_{\text{s}})$ is the number of configurations of the system with particle number N_{s} and $W_{\text{b}}(E_{\text{tot}} - E_{\text{s}}, N_{\text{tot}} - N_{\text{s}})$ is the number of configurations of the bath, with particle number $N_{\text{b}} = N_{\text{tot}} - N_{\text{s}}$, we may write the total number of configurations as:

$$W_{\text{tot}} = W_{\text{s}}(E_{\text{s}}, N_{\text{s}})\, W_{\text{b}}(E_{\text{tot}} - E_{\text{s}}, N_{\text{tot}} - N_{\text{s}}) \ .$$

where we assume the means of particle exchange does not significantly alter the counting of the configurations – so that we may assume the two bodies are statistically independent.

A physical realisation of this is two copies of the same volume of the sample, L^3. Initially they are separated from each other, but then they are brought together. The interaction between them is proportional to their shared surface, L^2. Across this surface both energy and particles

may be exchanged. However the number of atoms at the surface is negligible compared to the bulk. For example if the bulk is such that there are Avogadro's number of atoms in "each half", then the number at the interface is proportional to

$$\left(L^3\right)^{2/3} \propto \left(10^{23}\right)^{2/3} \simeq 10^{15} \ .$$

i.e. a considerably smaller number. Thus the "interaction energy", compared to the energy residing in the bulk of the two samples, is negligible.

Now we will exploit $N_\mathsf{s} \ll N_\mathsf{tot}$ to Taylor expand W_b, both in energy and particle number. However it is much more convenient to proceed (as in section 3.4) in a slightly roundabout way – as we understand derivatives of $S = k_\mathrm{B} \ln W$ more than derivatives of W itself.

$$\begin{aligned}
W_\mathsf{tot}\left(N_\mathsf{tot}, N_\mathsf{s}\right) &= W_\mathsf{s}(E_\mathsf{s}, N_\mathsf{s})\, W_\mathsf{b}(E_\mathsf{tot} - E_\mathsf{s}, N_\mathsf{tot} - N_\mathsf{s}) \\
&= W_\mathsf{s}(E_\mathsf{s}, N_\mathsf{s})\, \exp\left(\ln W_\mathsf{b}(E_\mathsf{tot} - E_\mathsf{s}, N_\mathsf{tot} - N_\mathsf{s})\right) \\
&= W_\mathsf{s}(E_\mathsf{s}, N_\mathsf{s}) \exp\left(\ln W_\mathsf{b}(E_\mathsf{tot}, N_\mathsf{tot}) - E_\mathsf{s} \left.\frac{\partial \ln W_\mathsf{b}}{\partial E}\right|_{E=E_\mathsf{tot}} - N_\mathsf{s} \left.\frac{\partial \ln W_\mathsf{b}}{\partial N}\right|_{N=N_\mathsf{tot}} + \cdots\right) \\
&\simeq W_\mathsf{s}(E_\mathsf{s}, N_\mathsf{s})\, W_\mathsf{b}(E_\mathsf{tot}, N_\mathsf{tot})\, \exp\left[-\left(\frac{\partial S_\mathsf{b}}{\partial E_\mathsf{b}}\right)_{V_\mathsf{b}} \frac{E_\mathsf{s}}{k_\mathrm{B}} - \left(\frac{\partial S_\mathsf{b}}{\partial N_\mathsf{b}}\right)_{V_\mathsf{b}} \frac{N_\mathsf{s}}{k_\mathrm{B}}\right] \\
\text{(Use Eq. (8.5).)} \quad &= W_\mathsf{s}(E_\mathsf{s}, N_\mathsf{s})\, W_\mathsf{b}(E_\mathsf{tot}, N_\mathsf{tot}) \exp\left(-\frac{(E_\mathsf{s} - \mu N_\mathsf{s})}{k_\mathrm{B} T_\mathsf{b}}\right) \ . \qquad (8.9)
\end{aligned}$$

Now add the contributions from each value of E_s and N_s, and since we treat N_s as a discrete variable now:

$$\begin{aligned}
W(N_\mathsf{tot}) &= \int \mathrm{d}E_\mathsf{s} \sum_{N_\mathsf{s}=0}^{\infty} W_\mathsf{tot}\left(E_\mathsf{tot} - E_\mathsf{s}, N_\mathsf{tot} - N_\mathsf{s}\right) \qquad (8.10) \\
&\simeq W_\mathsf{b}(E_\mathsf{tot}, N_\mathsf{tot}) \int \mathrm{d}E_\mathsf{s} \sum_{N_\mathsf{s}=0}^{\infty} W_\mathsf{s}(E_\mathsf{s}, N_\mathsf{s}) \exp\left(-\frac{(E_\mathsf{s} - \mu N_\mathsf{s})}{k_\mathrm{B} T_\mathsf{b}}\right) \\
\Rightarrow \quad W_\mathsf{s}(T, V) &= \frac{W(N_\mathsf{tot})}{W_\mathsf{b}(N_\mathsf{tot})} \simeq \int \mathrm{d}E_\mathsf{s} \sum_{N_\mathsf{s}=0}^{\infty} W_\mathsf{s}(E_\mathsf{s}, N_\mathsf{s}) \exp\left(-\frac{(E_\mathsf{s} - \mu N_\mathsf{s})}{k_\mathrm{B} T_\mathsf{b}}\right) \qquad (8.11) \\
&= \mathcal{Z}_\mathsf{s}(T, V, \mu) \ . \qquad (8.12)
\end{aligned}$$

where we have defined $\mathcal{Z}_\mathsf{s}(T, V)$, the *grand partition function*, for the system when in contact with a heat bath at temperature T and a particle bath, also at T, with chemical potential μ.

We can form (using the "strongly-peaked" arguments as before, in section 5 where we calculated the free energy) the *Grand potential*, $\Omega(T, V, \mu)$,

$$\begin{aligned}
\Omega(T, V, \mu) &= F(T, V, N) - \mu N \qquad (8.13) \\
&= -pV \\
&= -k_\mathrm{B} T \ln \mathcal{Z}(T, V, \mu) \ ,
\end{aligned}$$

where the last equality comes from comparison with Eq. (8.7). Notice there is no constant pressure ensemble corresponding to Ω as

$$\Omega(T,V,\mu) + pV = -pV + pV = 0 \ .$$

We may identify which thermodynamic functions are derivatives of the grand potential in the following manner, using: (i) the expression for the total differential of $\Omega(T,V,\mu)$; (ii) $\Omega = F - \mu N$; and (iii) the total differential for $F(T,V,N)$:

$$\begin{aligned} \mathrm{d}\Omega(T,V,\mu) &= \left(\frac{\partial\Omega}{\partial T}\right)_{V,\mu} \mathrm{d}T + \left(\frac{\partial\Omega}{\partial V}\right)_{\mu,T} \mathrm{d}V + \left(\frac{\partial\Omega}{\partial \mu}\right)_{T,V} \mathrm{d}\mu \qquad (8.14) \\ &= \mathrm{d}F - \mathrm{d}(\mu N) \\ &= S\mathrm{d}T - p\,\mathrm{d}V + \mu\mathrm{d}N - \mu\mathrm{d}N - N\mathrm{d}\mu \\ &= S\mathrm{d}T - p\,\mathrm{d}V - N\mathrm{d}\mu \qquad (8.15) \end{aligned}$$

Comparing Eq. (8.14) and Eq. (8.15) we see:

$$S = \left(\frac{\partial\Omega}{\partial T}\right)_{V,\mu} \qquad p = -\left(\frac{\partial\Omega}{\partial V}\right)_{\mu,T} \qquad N = -\left(\frac{\partial\Omega}{\partial \mu}\right)_{T,V} \ . \qquad (8.16)$$

8.3 Example of an ideal gas

We derived the Helmholtz free energy, $F(T,V,N)$, in section 5.1, Eq. (5.4):

$$F = -k_{\mathrm{B}}T\ln Z = -Nk_{\mathrm{B}}T\left\{\ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) + 1\right\} \ , \qquad (8.17)$$

where

$$v = V/N \quad \text{and} \quad \lambda_{\mathrm{T}} = \sqrt{\frac{\mathsf{h}^2}{2\pi m k_{\mathrm{B}}T}} \ .$$

We may now determine the chemical potential:

$$\begin{aligned} \mu = \left(\frac{\partial F}{\partial N}\right)_{T,V} &= -k_{\mathrm{B}}T\frac{\partial}{\partial N}N\left\{\ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) + 1\right\} \\ &= -k_{\mathrm{B}}T\left[\left\{\ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) + 1\right\} + N\frac{\lambda_{\mathrm{T}}^3}{v}\frac{\partial v}{\partial N}\frac{1}{\lambda_{\mathrm{T}}^3}\right] \\ &= -k_{\mathrm{B}}T\left[\left\{\ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) + 1\right\} + N\frac{\lambda_{\mathrm{T}}^3}{v}(-1)\frac{V}{N^2}\frac{1}{\lambda_{\mathrm{T}}^3}\right] \\ &= -k_{\mathrm{B}}T\left[\left\{\ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) + 1\right\} - 1\right] \\ \Rightarrow \quad \mu &= -k_{\mathrm{B}}T\ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) \ . \qquad (8.18) \end{aligned}$$

Note this, as before, makes sense when the argument of the ln is significantly greater than unity, so there are a large number of states available for each particle and the gas is in classical limit. Otherwise the full quantum treatment is needed.

The interpretation is that since $\mu < 0$ the chemical potential is dominated by the entropy per particle, remembering that $F = U - TS$. Note $\mu \to -\infty$ as $T \to \infty$, consistently, as the number of states available diverges as $\lambda_{\mathrm{T}} \to 0$. Conversely, dominance by the internal energy only occurs in the quantum regime and then only for fermions, as we shall see.

Important. An application of the ideal gas chemical potential to the degree of ionisation of a hydrogen atomic gas (Saha's equation, which is important in astrophysics) is in problem 4.1.

Let us turn to the grand potential, Ω, for the ideal gas. We will determine Ω in three different ways, providing complementary perspectives.

Firstly, we may use Eq. (8.13) to deduce the grand potential:

$$\begin{aligned}
\Omega &= F - \mu N \\
\text{(Use Eq. (8.17) and Eq. (8.18).)} \quad &= -Nk_{\mathrm{B}}T\left\{\ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) + 1\right\} - (-1)Nk_{\mathrm{B}}T\ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) \\
&= -Nk_{\mathrm{B}}T \\
\text{(Use equation of state.)} \quad &= -pV \ .
\end{aligned}$$

A faster, but maybe less satisfying, route is:

$$\begin{aligned}
\Omega &= F - \mu N \\
&= F - G \\
&= -pV \ .
\end{aligned}$$

Finally let us check this by direct calculation of Ω from $\mathcal{Z}(T, V, \mu)$.

The remainder of this section is non-examinable

Remember the expression for the area, $W(E, V, N)$, of the hypersurface in classical phase space of energy E with N particles each inhabiting the same volume V in d-dimensions. The appropriate scaling by powers of $\hbar$ and the $1/N!$ from the identical nature of the particles are both included. Then, with Hamiltonian $\mathcal{H}$:

$$W(E_{\mathsf{s}}, V, N_{\mathsf{s}}) = \sum_{N=0}^{\infty} \frac{\delta_{N_{\mathsf{s}},N}}{N!\ \mathsf{h}^{dN}} \int \prod_{n=1}^{N} \mathrm{d}\mathbf{q}_n \mathrm{d}\mathbf{p}_n \delta\left[E_{\mathsf{s}} - \mathcal{H}(\mathbf{q}_1, \cdots, \mathbf{q}_N; \mathbf{p}_1, \cdots, \mathbf{p}_N)\right] \ .$$

(Note: there is the awkward case of $N = 0$. There we define the phase space integral to be unity, i.e. there is one state with no particles.)

We may now substitute this expression into the grand partition function, Eq. (8.12), using the shorthand introduced in section 1 Q_N, P_N and $\mathrm{d}Q_N\mathrm{d}P_N$ to denote the collections of

coordinates, momenta and the combined element of integration. We drop the subscript b to denote bath now:

$$\begin{aligned}
\mathcal{Z}_{\mathsf{s}}(T,V,\mu) &= \int \mathrm{d}E_{\mathsf{s}} \sum_{N_{\mathsf{s}}=0}^{\infty} W_{\mathsf{s}}(E_{\mathsf{s}},N_{\mathsf{s}}) \exp\left(-\frac{(E_{\mathsf{s}}-\mu N_{\mathsf{s}})}{k_{\mathrm{B}}T}\right) \\
&= \int \mathrm{d}E_{\mathsf{s}} \sum_{N_{\mathsf{s}}=0}^{\infty} \underbrace{\sum_{N=0}^{\infty} \delta_{N_{\mathsf{s}},N} \int \frac{\mathrm{d}Q_N \mathrm{d}P_N}{N!\,\mathsf{h}^{dN}} \delta\left[E_{\mathsf{s}} - \mathcal{H}(Q_N;P_N)\right]}_{W_{\mathsf{s}}(E_{\mathsf{s}},N_{\mathsf{s}})} \exp\left(-\frac{(E_{\mathsf{s}}-\mu N_{\mathsf{s}})}{k_{\mathrm{B}}T}\right) \\
&= \sum_{N=0}^{\infty} \int \frac{\mathrm{d}Q_N \mathrm{d}P_N}{N!\,\mathsf{h}^{dN}} \exp\left(-\frac{[\mathcal{H}(Q_N,P_N)-\mu N]}{k_{\mathrm{B}}T}\right) \qquad (\delta\text{'s do sum/integral.}) \ .
\end{aligned}$$

At this point we need to make explicit the system being an ideal gas. The Hamiltonian is then only kinetic energy

$$\mathcal{H}(\mathbf{q}_1,\cdots,\mathbf{q}_N;\mathbf{p}_1,\cdots,\mathbf{p}_N) = \sum_{n=1}^{N} \frac{\mathbf{p}_n^2}{2m} \ .$$

Then we may separate the terms in the exponent using the "exponential of a sum is the product of exponentials" result from section 7.5, Eq. (7.2), with $1 \le n \le \infty$, allowing us to write:

$$\begin{aligned}
\mathcal{Z}_{\mathsf{s}}(T,V,\mu) &= \sum_{N=0}^{\infty} \frac{1}{N!} \left\{ \prod_{n=1}^{N} \int \frac{\mathrm{d}\mathbf{q}_n \mathrm{d}\mathbf{p}_n}{\mathsf{h}^d} \exp\left(-\beta \frac{\mathbf{p}_n^2}{2m}\right) \right\} \mathrm{e}^{\beta\mu N} \\
&= \sum_{N=0}^{\infty} \frac{1}{N!} \left\{ \prod_{n=1}^{N} Z_n \right\} \mathrm{e}^{\beta\mu N} \\
\text{(Each term in product identical.)} \qquad &= \sum_{N=0}^{\infty} \frac{Z_1^N}{N!} \mathrm{e}^{\beta\mu N}
\end{aligned}$$

Here, all the separated single-particle partition functions are the same

$$Z_1 = \frac{V}{\mathsf{h}^d} \int \mathrm{d}\mathbf{p} \exp\left(-\beta \frac{\mathbf{p}^2}{2m}\right) \ .$$

Now we may perform the sum over N – it is the Taylor series for an exponential:

$$\begin{aligned}
\mathcal{Z}(T,V,\mu) &= \exp\left(Z_1 \mathrm{e}^{\beta\mu}\right) \\
\Rightarrow \quad \Omega(T,V,\mu) &= -k_{\mathrm{B}}T \ln \mathcal{Z}(T,V,\mu) \\
&= -k_{\mathrm{B}}T \left(Z_1 \mathrm{e}^{\beta\mu}\right) \\
&= -k_{\mathrm{B}}TV\lambda_{\mathrm{T}}^{-3} \frac{\lambda_{\mathrm{T}}^3}{v} \\
&= -k_{\mathrm{B}}TN \ ,
\end{aligned}$$

in agreement.